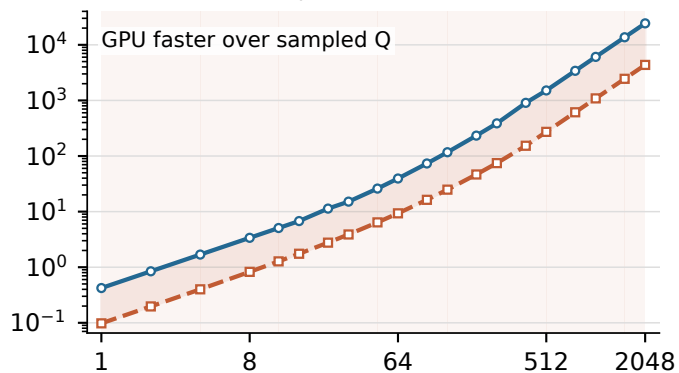


Prefill attention crossover · LLAMA3.1-8B

GPU PIM PIM lower cost GPU lower cost

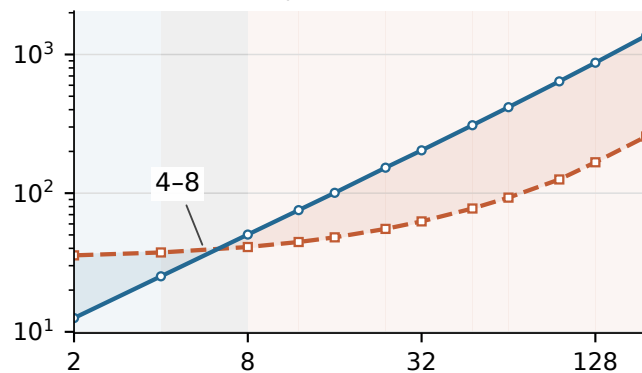
(a) Without MQ · Cache: 0

NVLink 3 · 300 GB/s



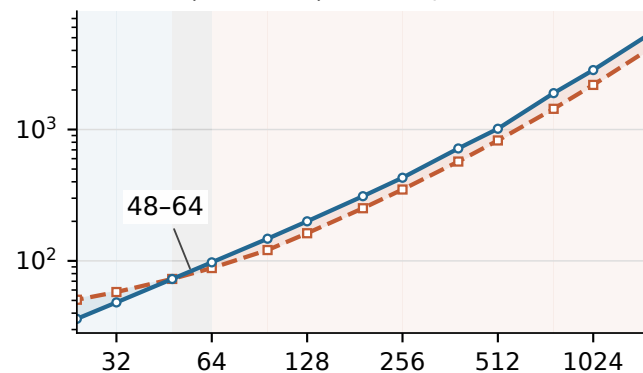
(b) Without MQ · Cache: 1,024

NVLink 3 · 300 GB/s



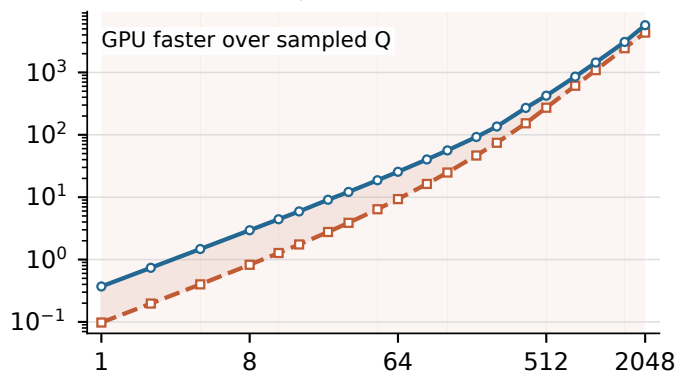
(c) With MQ · Cache: 1,024

NVLink 4 (maximum) · 450 GB/s



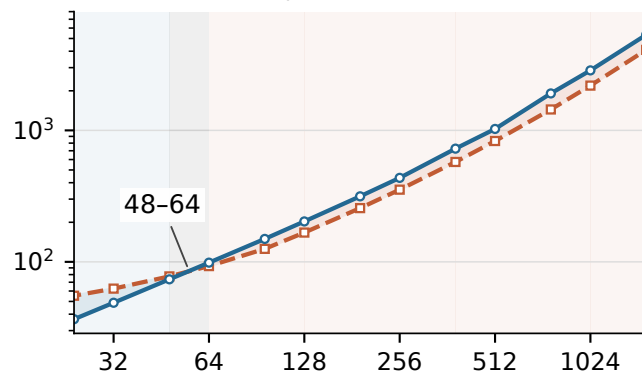
(d) With MQ · Cache: 0

NVLink 3 · 300 GB/s



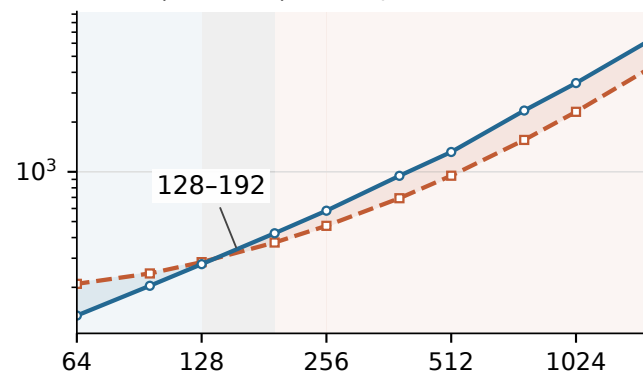
(e) With MQ · Cache: 1,024

NVLink 3 · 300 GB/s



(f) With MQ · Cache: 1,024

PCIe 4 (minimum) · 32 GB/s



New query tokens, Q